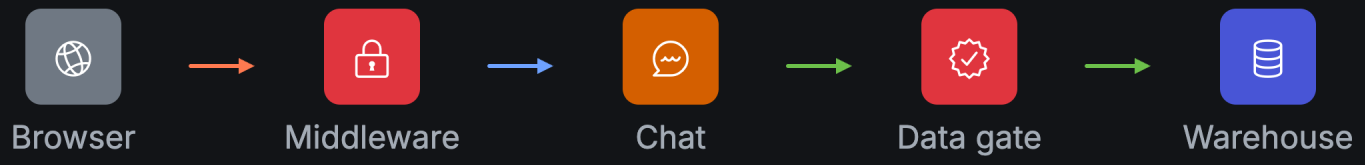
ARCHITECTURE OVERVIEW

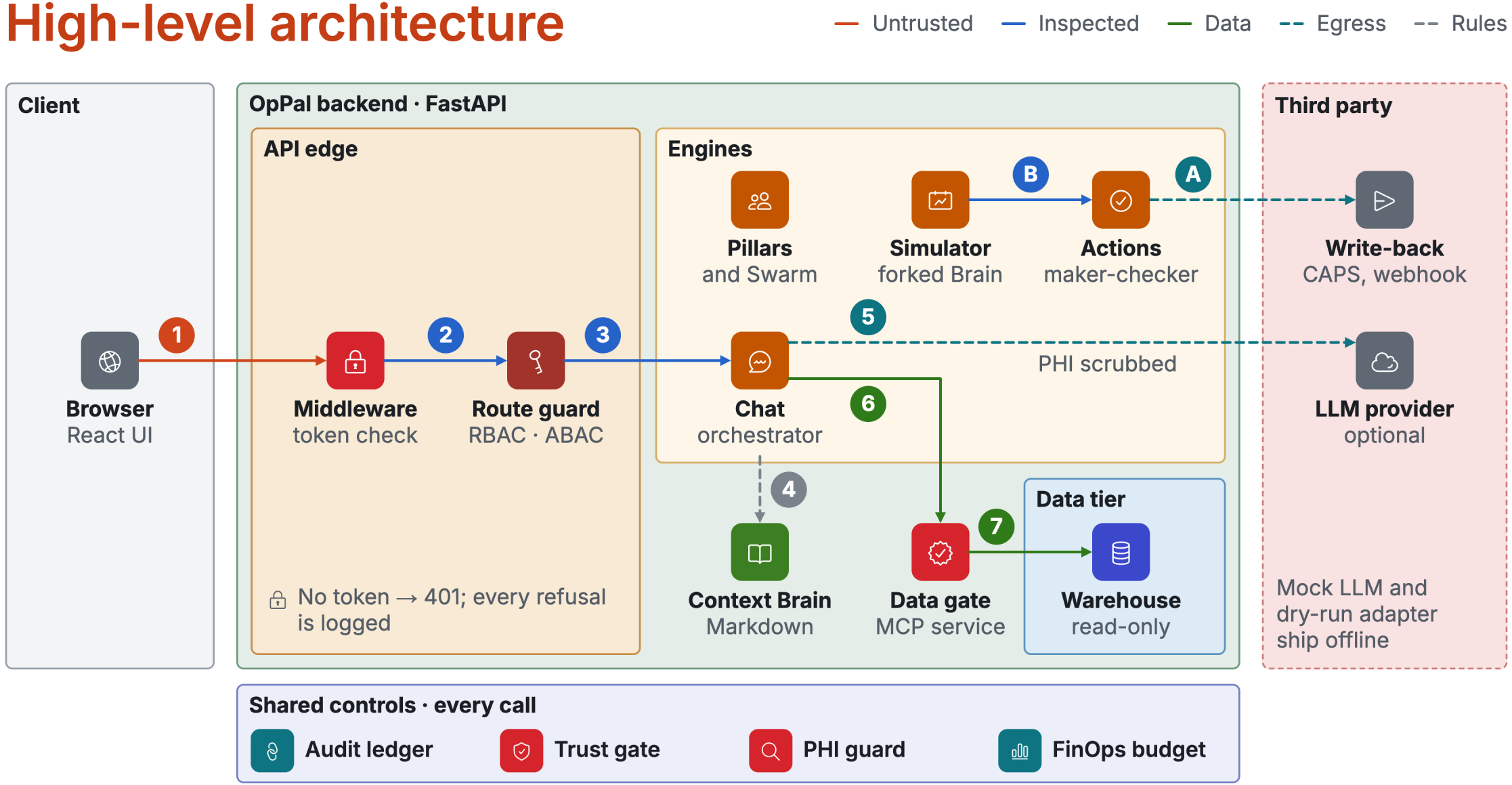
OpPal, high-level architecture

A plain-English question becomes a scoped, read-only, audited answer. Any write needs two signatures.



Build v14.14.0 · branch langchain-chat-agent · October 2026

High-level architecture



One request, hop by hop

Step	Hop	Control at the crossing
1	Browser → Middleware	HTTPS with a bearer token; no token → 401; 120 requests a minute
2	Middleware → Route guard	Principal re-resolved from the token; route permission checked
3	Route guard → Chat	RBAC and ABAC: no line-of-business grant → degraded answer
4	Chat → Context Brain	Question grounded in the domain's metrics, rules and vocabulary
5	Chat → LLM provider	PHI scrubbed first; budget and kill switch; mock provider offline
6	Chat → Data gate	Scope pushed down; one read-only statement; execute() is final
7	Data gate → Warehouse	Row-capped query over only the caller's lines of business
A	Actions → Write-back	Two distinct officers sign and the trust gate clears; dry-run default
B	Simulator → Actions	Runs on a forked Brain; promotion proposes, never writes

Each hop is appended to the answer's audit trail, which the UI draws as the journey strip, and every call is hash-chained into the ledger.

Eight stages behind every chat answer

1 Access control

RBAC and ABAC. No line-of-business grant → degraded answer.

2 Semantic grounding

Graph-RAG maps the question onto the Brain's metrics.

3 Intent routing

LLM or rule router. An out-of-scope filter is refused here.

4 Cache probe

A hit replays the stored provenance receipt.

5 SQL generation

LLM agent, or a governed template when it falls short.

6 Validate & execute

One read-only statement through the MCP data gate.

7 Root cause & trust

0.35 grounding, 0.30 gates, 0.20 sufficiency, 0.15 lineage.

8 Visualize & narrate

Chart, narrative and provenance receipt, then cached.

0.85 **Hard trust gate.** Below it the SQL is withheld and the question goes to the review queue.

Fallback ladder. LLM-written SQL that ABAC refuses, that lacks the metric column or fails a gate is retried on a governed template. A model can never widen scope.

Writes need two signatures and leave a hash

1 Simulate

Overrides land on a deep copy of the Brain. Promotion proposes an action; it never writes.

2 Approve twice

Two distinct compliance officers sign. The maker can't be the checker.

3 Execute

A governed adapter writes: dry-run by default, CAPS or webhook when configured.

4 Reverse

Only by an append-only compensating entry that cites the original.

One ledger, two chains

Every step is hashed into audit_log.jsonl: a per-call chain (v1) beside a SHA-256 chain with a monotonic seq and a persisted head (v2).

Protected keys

The dual-control threshold, severity triggers and trust gate are hardcoded, so no single officer can lower them and sign alone.